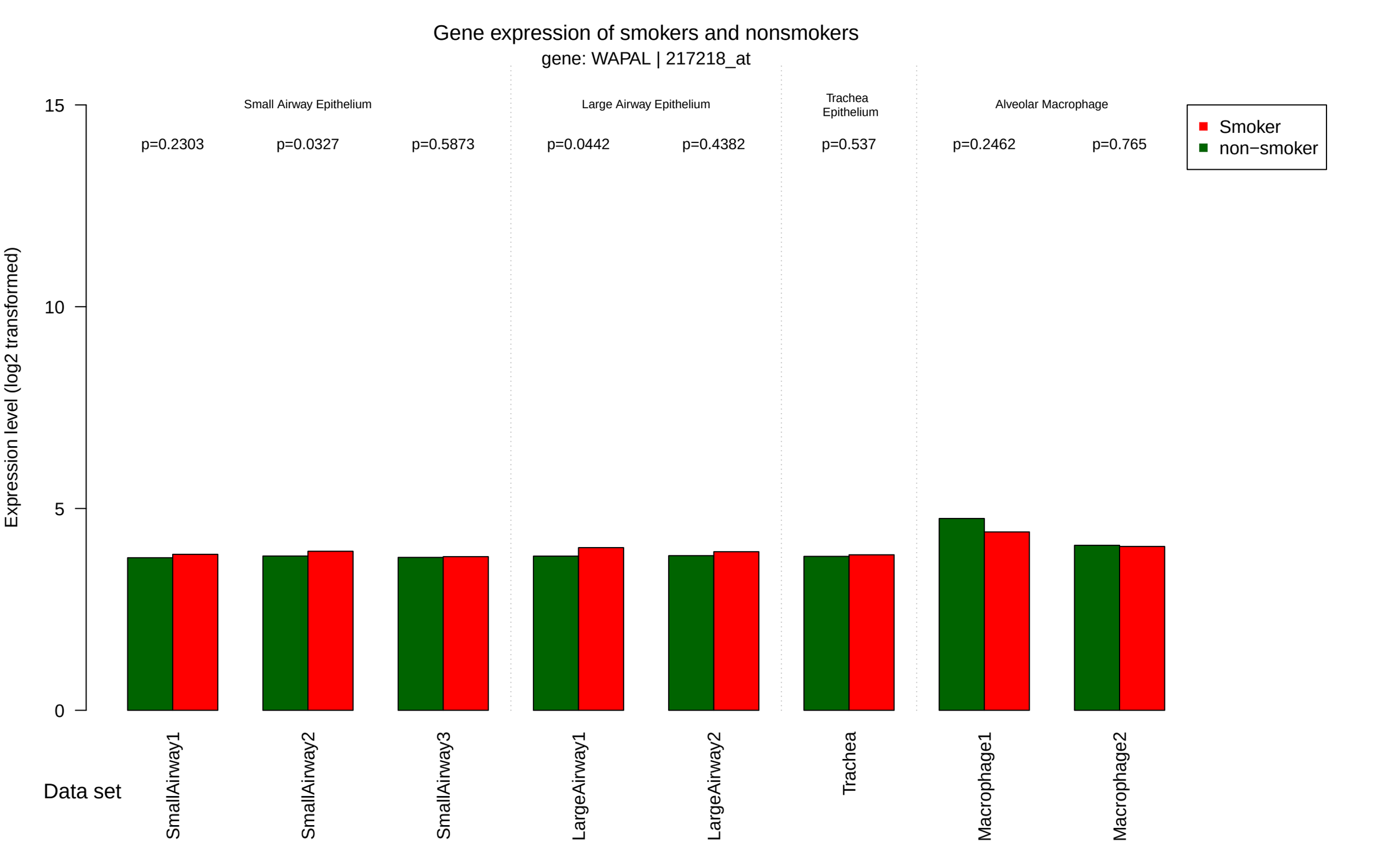
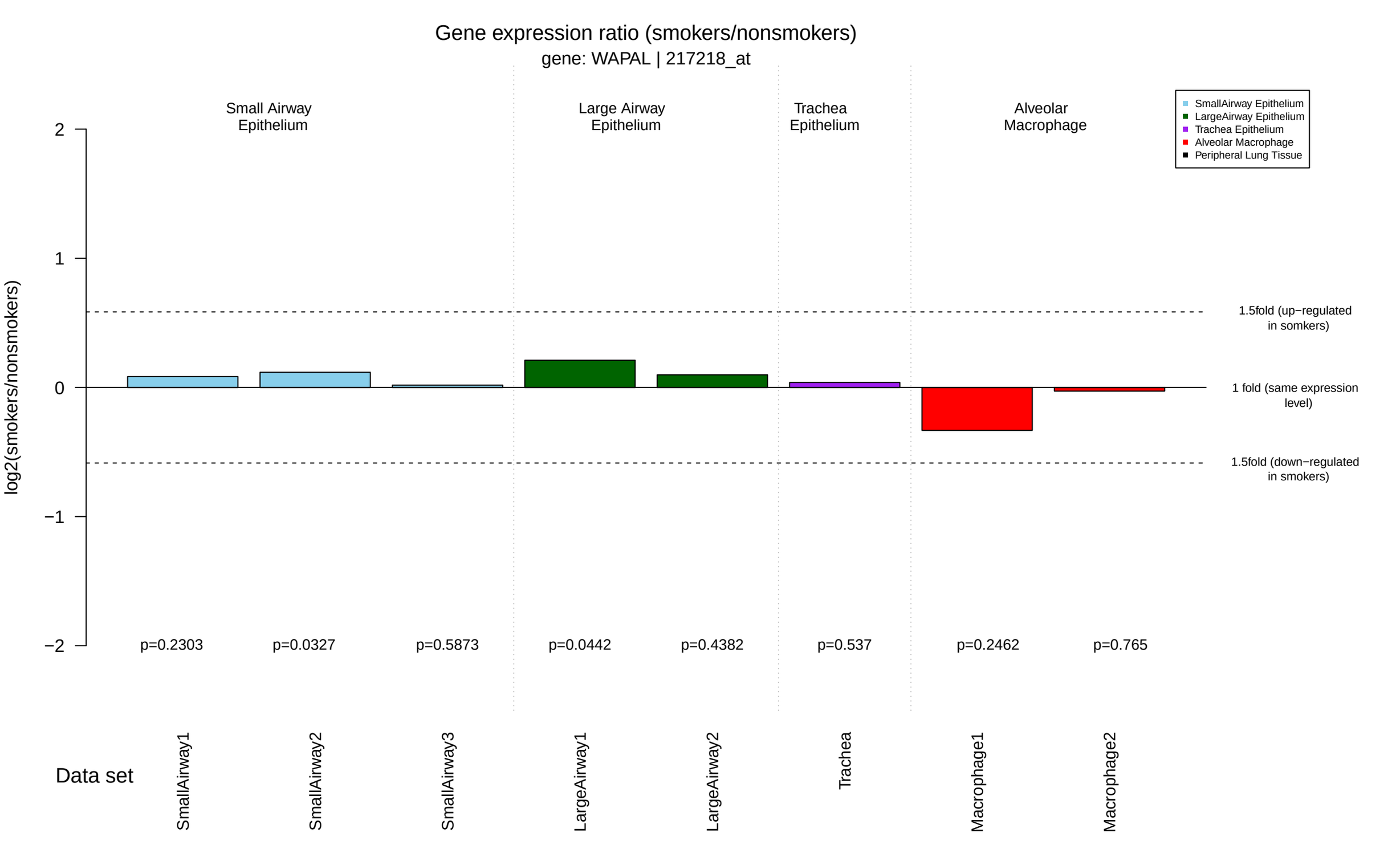
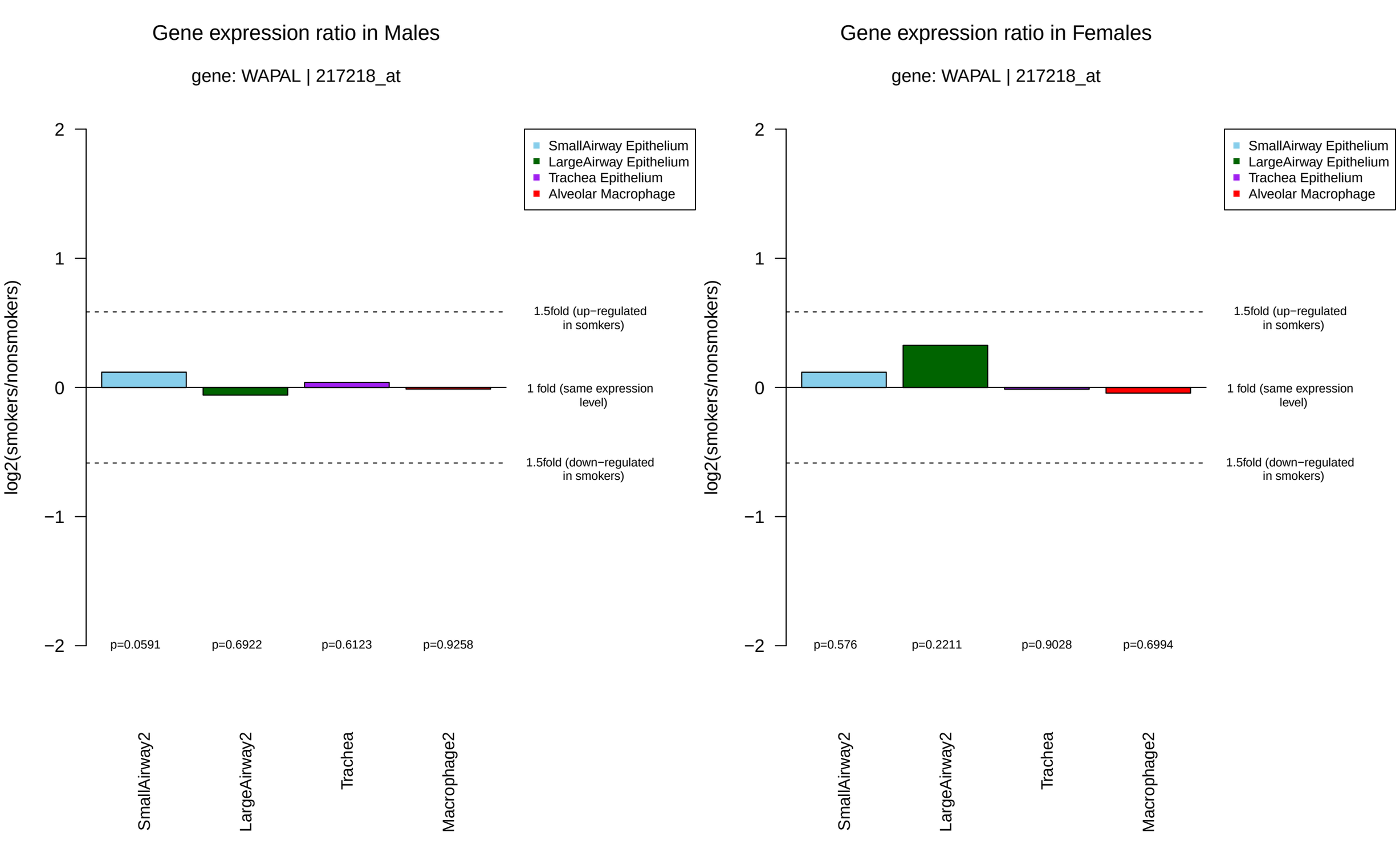
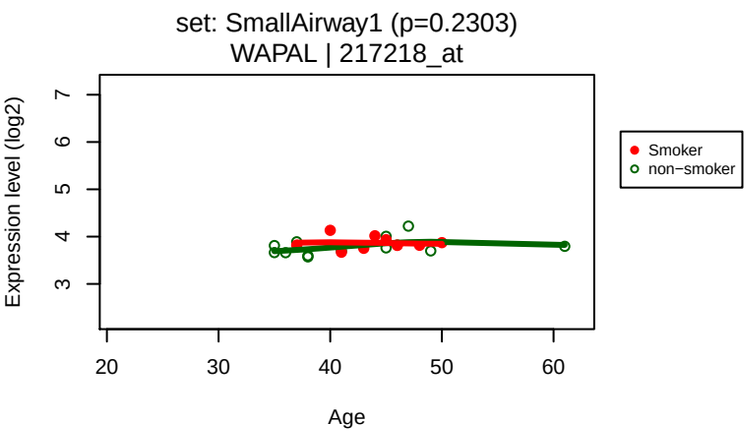
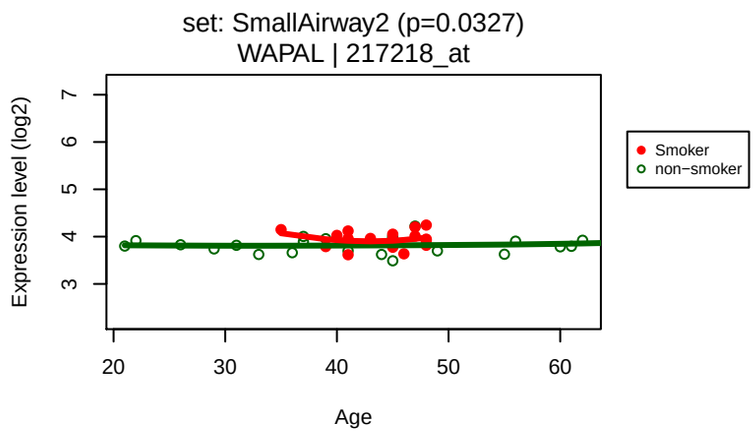
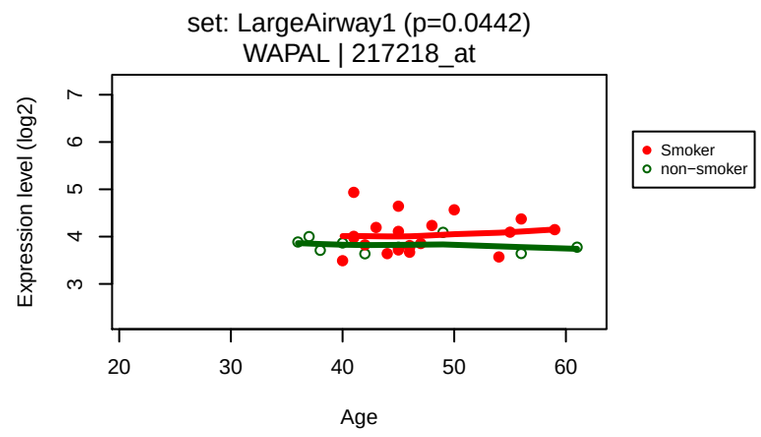

Gene expression of smokers and nonsmokers
gene: WAPAL | 217218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2303
p=0.0327
p=0.5873
p=0.0442
p=0.4382
p=0.537
p=0.2462
p=0.765
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

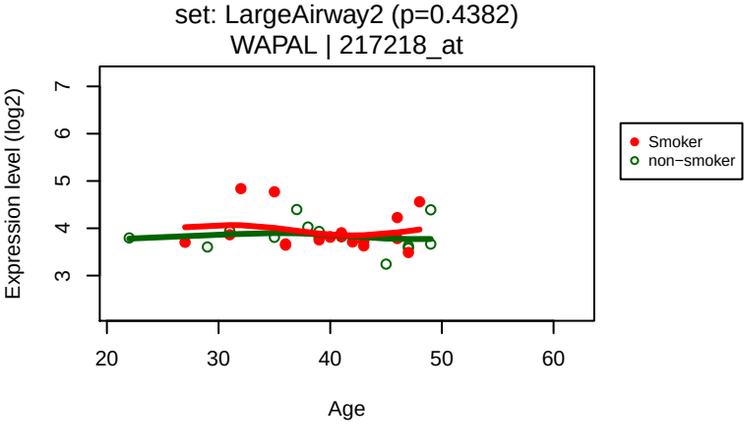

Gene expression ratio (smokers/nonsmokers)
gene: WAPAL | 217218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2303
p=0.0327
p=0.5873
p=0.0442
p=0.4382
p=0.537
p=0.2462
p=0.765
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

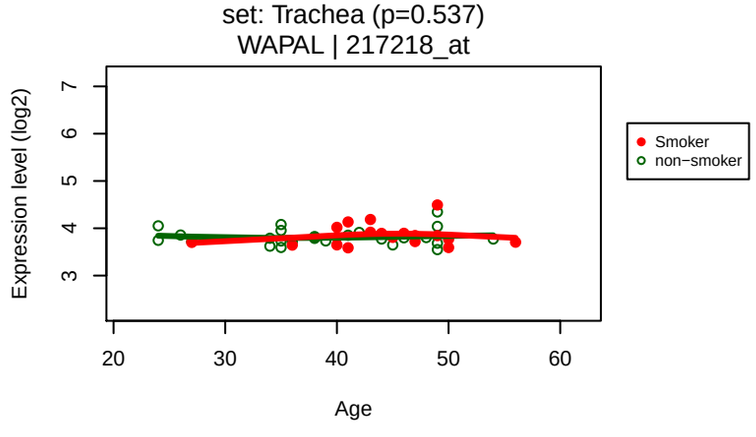

Gene expression ratio in Males
gene: WAPAL | 217218_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0591
p=0.6922
p=0.6123
p=0.9258
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WAPAL | 217218_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.576
p=0.2211
p=0.9028
p=0.6994
SmallAirway2
LargeAirway2
Trachea
Macrophage2

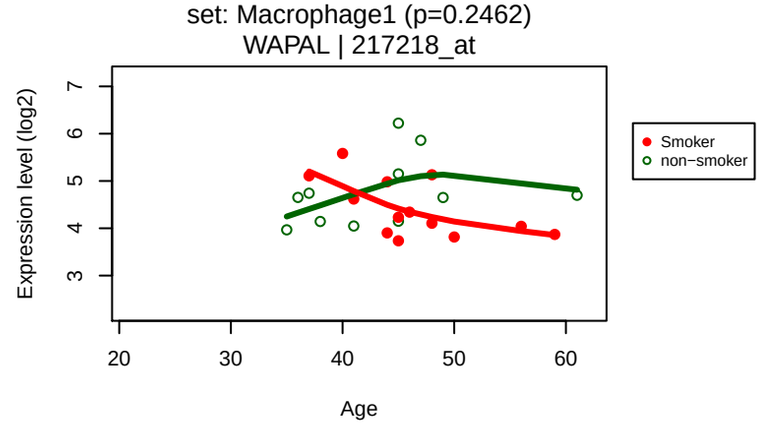

set: SmallAirway1 (p=0.2303)
WAPAL | 217218_at
Expression level (log2)
Age
Smoker
non−smoker

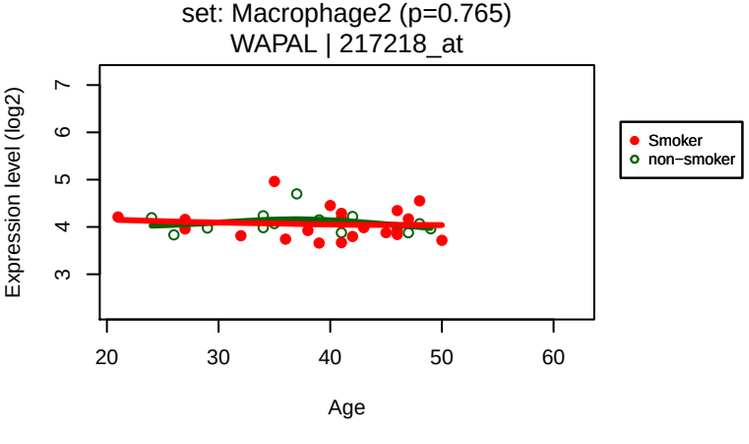

set: SmallAirway2 (p=0.0327)
WAPAL | 217218_at
Expression level (log2)
Age
Smoker
non−smoker

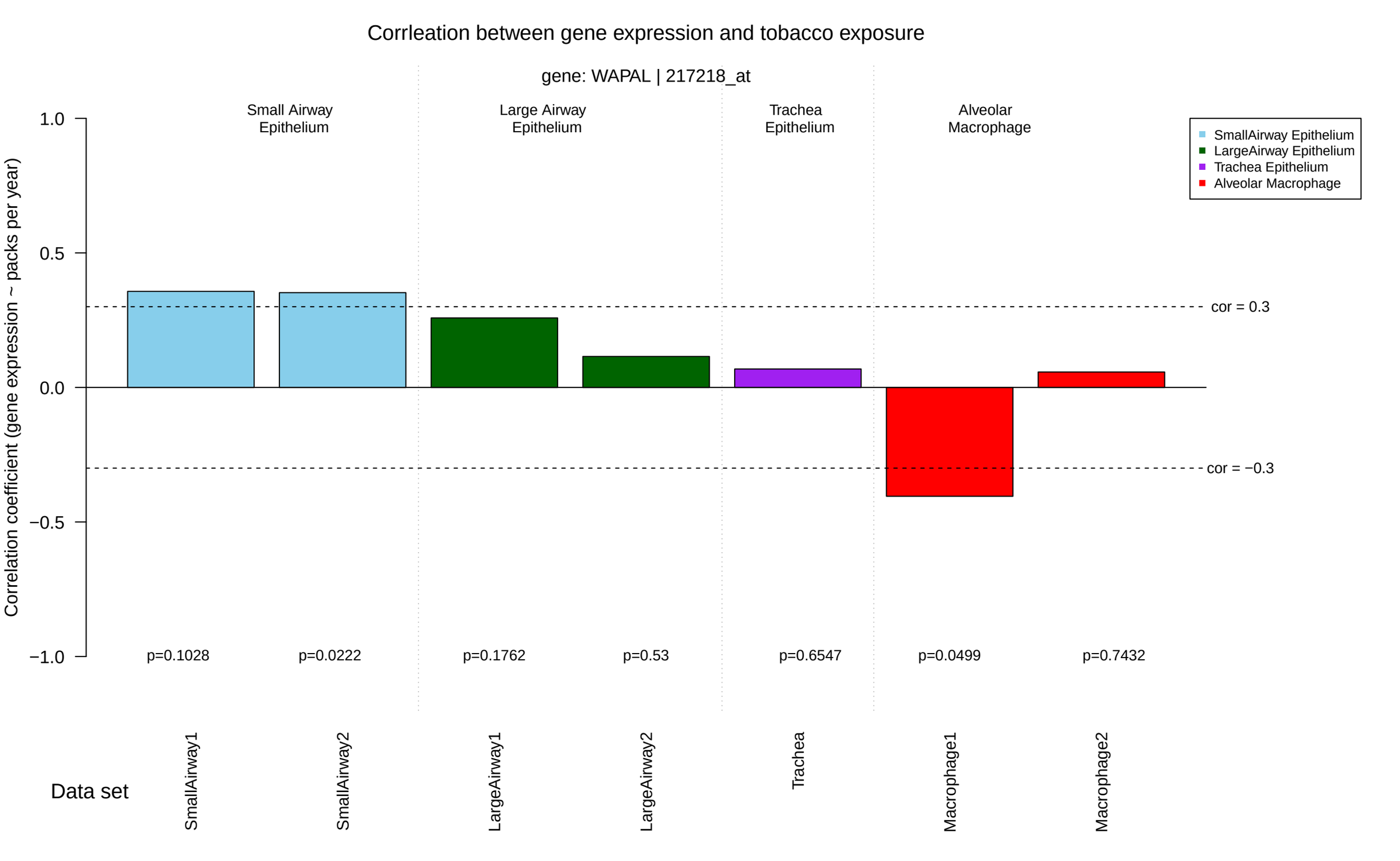

set: LargeAirway1 (p=0.0442)
WAPAL | 217218_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4382)
WAPAL | 217218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.537)
WAPAL | 217218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2462)
WAPAL | 217218_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.765)
WAPAL | 217218_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WAPAL | 217218_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1028
p=0.0222
p=0.1762
p=0.53
p=0.6547
p=0.0499
p=0.7432
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2